

Lab 11 - Expected output reference

All Pod IPs, ClusterIPs, ReplicaSet hashes, EndpointSlice name suffixes and ages are allocated at runtime and will differ on your cluster. Compare the **shape** and the **relationships** between values, not the literal digits.

bash verification/checks.sh - clean run

```
== Lab 11 verification: Services, Endpoints and Cluster DNS ==
PASS namespace kcna-lab11 exists
PASS ConfigMap quote-api-content matches data/quote-api-index.html byte-for-byte
PASS deployment quote-api has 3/3 ready replicas
PASS service quote-api is ClusterIP with port 80
PASS service quote-api targetPort is the named port "http"
PASS EndpointSlice resolved the named port to 8080
PASS EndpointSlice holds 3 ready addresses
PASS EndpointSlice addresses match the quote-api Pod IPs exactly
PASS dns fixture: quote-api resolves
PASS dns fixture: quote-api.kcna-lab11 resolves
PASS dns fixture: quote-api.kcna-lab11.svc resolves
PASS dns fixture: quote-api.kcna-lab11.svc.cluster.local resolves
PASS dns fixture: quote-api.default.svc.cluster.local correctly does NOT resolve
PASS dns fixture: kubernetes.default.svc.cluster.local resolves
PASS HTTP GET through the Service returns the Meridian Freight page
-- 15 passed, 0 failed --
```

Exit status 0. Any FAIL line prints the expected and observed values inline.

Key intermediate outputs

The Service (Step 4)

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
quote-api	ClusterIP	10.96.114.23	<none>	80/TCP	6s

EXTERNAL-IP is <none> and stays <none>. A ClusterIP Service is reachable only from inside the cluster - that is not a defect.

The EndpointSlice (Step 5)

NAME	ADDRESSTYPE	PORTS	ENDPOINTS	AGE
quote-api-t8hkb	IPv4	8080	10.244.0.14,10.244.0.15,10.244.0.16	45s

The two load-bearing facts:

Column	Value	Why it matters
PORTS	8080	The Service says targetPort: http; the slice records the resolved number. If this said <unset>, the named port did not resolve.
ENDPOINTS	three Pod IPs	Identical set to <code>kubectl get pods -l app=quote-api -o wide</code> . If this is empty, the selector matched nothing.

Per-endpoint detail:

```
10.244.0.14■true■quote-api-6c9b4f7d55-7lqm2
10.244.0.15■true■quote-api-6c9b4f7d55-jd8vx
10.244.0.16■true■quote-api-6c9b4f7d55-x2t9k
```

Column 2 is conditions.ready. Only true endpoints receive traffic.

Resolver configuration (Step 6)

```
search kcna-lab11.svc.cluster.local svc.cluster.local cluster.local
nameserver 10.96.0.10
options ndots:5
```

10.96.0.10 is the kube-dns Service ClusterIP in kube-system; on a default kind cluster it is the tenth address of the service CIDR.

DNS answer (Step 7)

```
Server:■■10.96.0.10
Address:■10.96.0.10:53

Name:■quote-api.kcna-lab11.svc.cluster.local
Address: 10.96.114.23
```

Known BusyBox behaviour - two separate artefacts, both observed:

1. busybox:1.36 also issues an AAAA query. On an IPv4-only kind cluster you may

additionally see *** Can't find ...: No answer and a non-zero exit code, *even for a name that resolved correctly over IPv4*.

2. **BusyBox nslookup does not apply the search list to a name that already

contains a dot.** nslookup quote-api.kcna-lab11 and nslookup quote-api.kcna-lab11.svc therefore report No answer on a cluster where those names resolve perfectly for real clients, because nslookup never appends svc.cluster.local / cluster.local.

Neither is a Service fault. verification/checks.sh therefore resolves each fixture with ping -c 1 -W 1 and reads only the address in the PING <name> (<ip>) header - ping uses the standard resolver and honours /etc/resolv.conf. ICMP replies are ignored, because a ClusterIP is a kube-proxy DNAT rule and is under no obligation to answer ping:

```
$ kubectl exec -n kcna-lab11 dns-client -- ping -c 1 -W 1 quote-api.kcna-lab11
PING quote-api.kcna-lab11 (10.96.114.23): 56 data bytes

--- quote-api.kcna-lab11 ping statistics ---
1 packets transmitted, 0 packets received, 100% packet loss
```

Resolution succeeded (address in parentheses); packet loss is expected and is not part of the assertion. A name that genuinely does not resolve looks completely different:

```
$ kubectl exec -n kcna-lab11 dns-client -- ping -c 1 -W 1 quote-api.default.svc.cluster.local
ping: bad address 'quote-api.default.svc.cluster.local'
command terminated with exit code 1
```

Cross-namespace negative (Step 7)

```
Server:██10.96.0.10
Address:██10.96.0.10:53

*** Can't find quote-api.default.svc.cluster.local: No answer

command terminated with exit code 1
```

This failure is correct and expected. A Service record only exists under its own namespace.

Failure-injection reference (Section 6)

Symptom:

```
wget: can't connect to remote host (10.96.203.77): Connection refused
command terminated with exit code 1
```

Empty slice:

NAME	ADDRESSTYPE	PORTS	ENDPOINTS	AGE
quote-api-broken-4zq6n	IPv4	<unset>	<unset>	40s

kubectl describe service quote-api-broken - the decisive two lines:

```
Selector:          app=quote-apis
Endpoints:
```

Root-cause proof:

```
$ kubectl get pods -n kcna-lab11 -l app=quote-apis
No resources found in kcna-lab11 namespace.
```

After the patch, the slice repopulates without any Pod restart:

NAME	ADDRESSTYPE	PORTS	ENDPOINTS	AGE
quote-api-broken-4zq6n	IPv4	8080	10.244.0.14,10.244.0.15,10.244.0.16	2m5s

Environment caveats recorded for this lab

Caveat	Effect on this lab
busybox:1.36 nslookup also queries AAAA	Extra No answer line and a non-zero exit for names that resolve fine over IPv4
busybox:1.36 nslookup ignores the search list for dotted partial names	quote-api.kcna-lab11 and quote-api.kcna-lab11.svc report No answer under nslookup but resolve correctly under ping, wget and any normal application resolver. Checks use ping's resolution line
A ClusterIP does not answer ICMP	ping shows 100% packet loss for a healthy Service. Only the resolved address in the PING <name> (<ip>) header is meaningful
v1 Endpoints deprecated in Kubernetes v1.33	kubect! get endpoints still works today but is not the API to build on; endpointslices is used throughout
kube-proxy in iptables mode REJECTs a ClusterIP with no endpoints	Produces *Connection refused* immediately. On a cluster in IPVS mode, or behind some CNIs, the same fault can present as a timeout instead
ClusterIP is cluster-internal only	Nothing in this lab is reachable from your laptop without kubect! port-forward; that is by design and is explored in Lab 12